FabCon Atlanta
2026

Module 4 – OneLake Security

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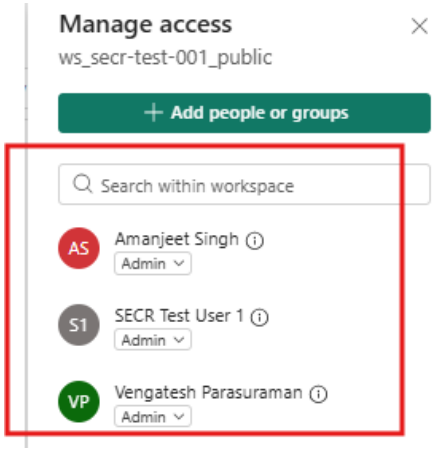
Overview

For this exercise, you will be using **ws**_<*user*>_**public**. Before you commence this module, please ensure that only the following users have workspace access assigned to _*public* assigned to you. Besides your account, just the instructors should have administrator access to your workspace.

Please DO NOT remove admin access for administrators from your workspace.

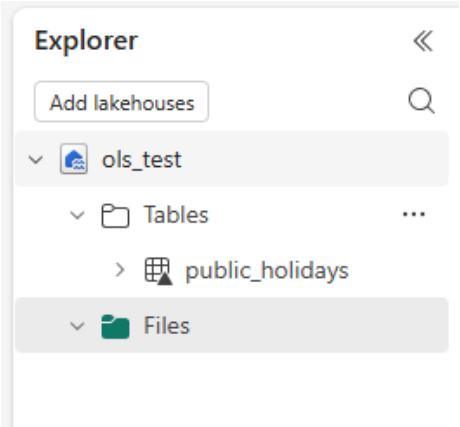


In this module, you will opt in to OneLake security and configure row-level (RLS) and column-level security (CLS) policies. You will then assign the policy to a viewer account (referred to as **test_viewer** in this document) to validate the effect of the OneLake security policy. Your instructors will share viewer account details for you to test.

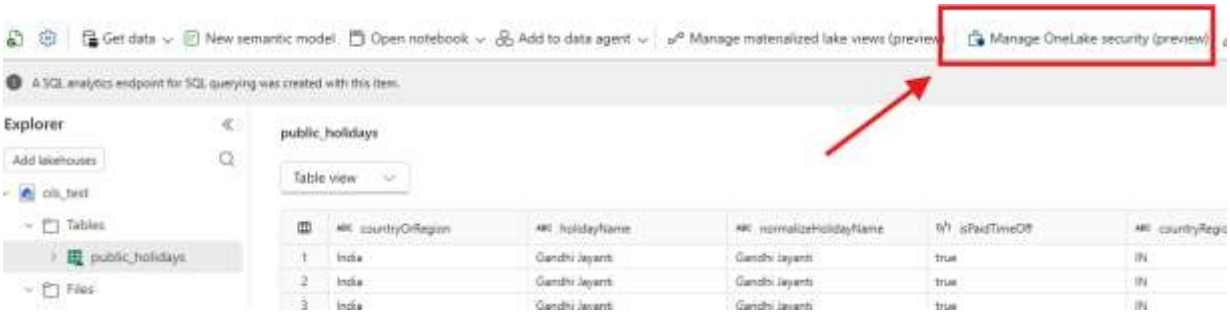
Step -1 Opt-in to OneLake security via a lakehouse item

Opt-in for Fabric OneLake security.

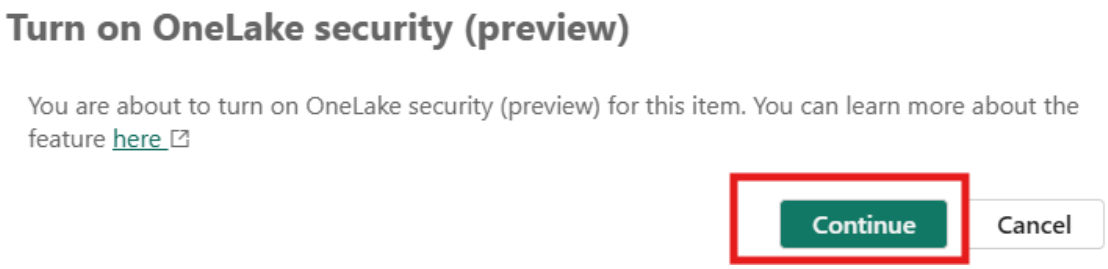
1. Log in to Fabric and navigate to workspace **ws**_<*user*>_**public**.
2. Locate **public_holidays_lh** lakehouse. It is preloaded with a sample **public holidays** dataset.



3. Select **Manage OneLake security (preview)**. This is a one-time, irreversible opt-in. After opt-in, only users assigned OneLake security roles can read or write data; all others have no access.

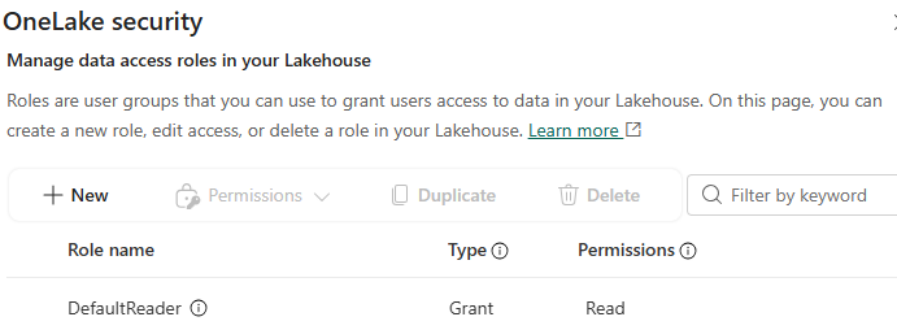


4. Select **Continue** when prompted to switch-on OneLake security for this lakehouse.

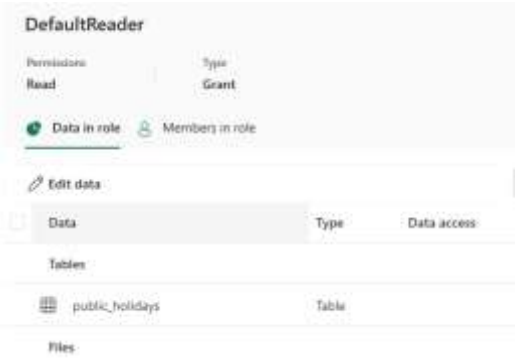


5. After opt-in, the OneLake security configuration experience appears with a pre-provisioned **DefaultReader** role.

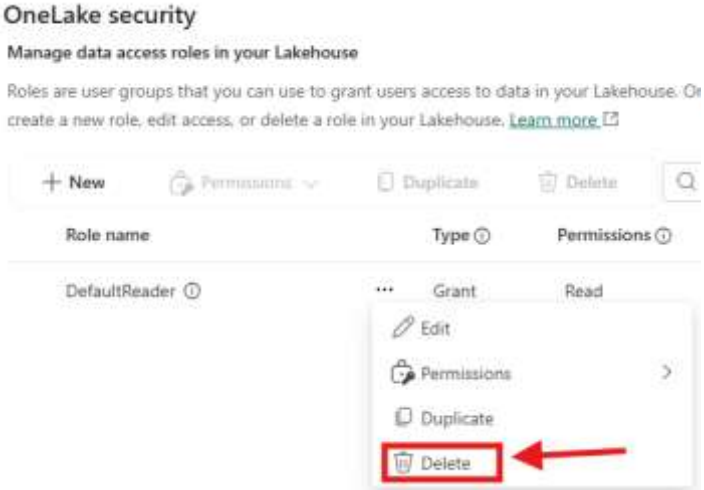
   The **DefaultReader** role is a built-in OneLake security role created automatically for every new lakehouse or data item in Microsoft Fabric. Its primary purpose is to provide a default level of read access to users who have the ReadAll permission, allowing them to view data without requiring manual role assignment. This ensures that users can access data immediately after a lakehouse is created, facilitating a smooth opt-in experience.



6. Select the role to view its preconfigured settings, which include all tables and folders with read access granted to all workspace members.



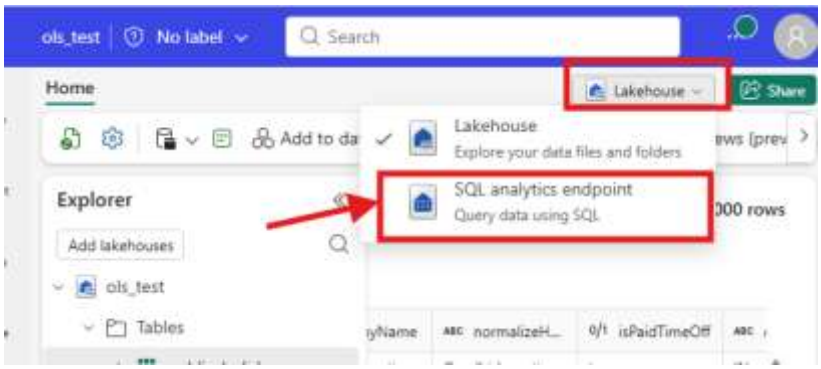
7. Delete **DefaultReader** role.



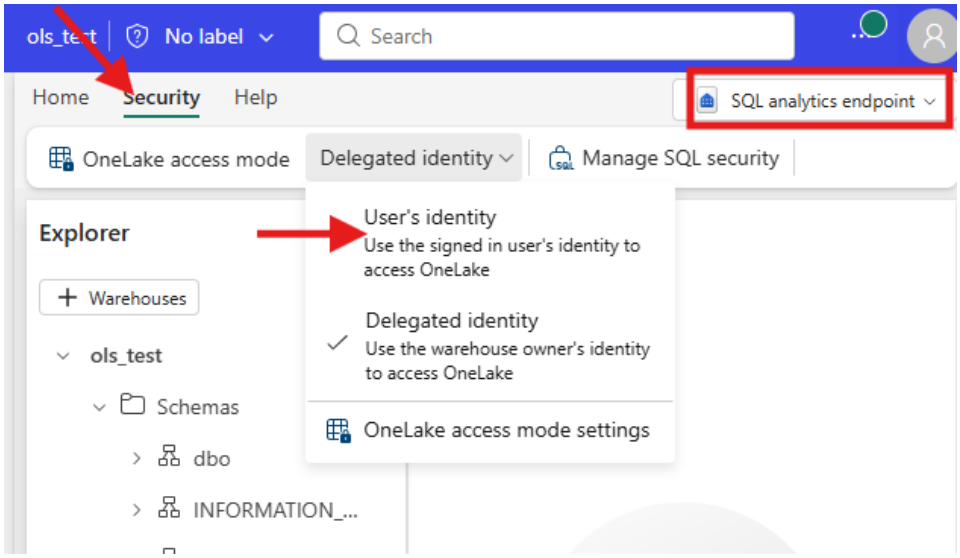
Step 2 – Enable User Identity mode for SQL analytics endpoint

SQL analytics endpoint (SQL AEP) requires an additional step to opt-in for OneLake security. By default, SQL analytics endpoint uses **Delegated Identity**. To honour OneLake security policies, you must switch it to **User Identity** mode.

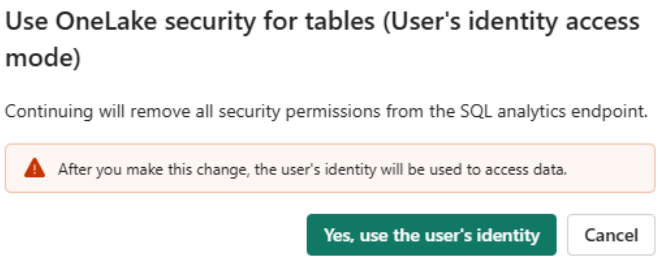
8. From the lakehouse user experience, switch to SQL analytics endpoint experience by selecting SQL **analytics endpoint** from the dropdown menu.



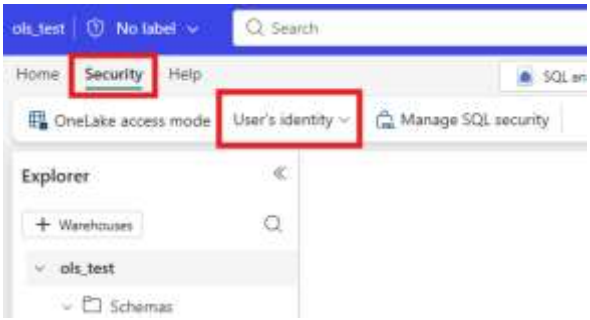
9. In SQL analytics endpoint experience, select **Security** and then change mode to **User's Identity.**



10. When prompted, select **Yes, use the user's identity.**



11. If applied correctly, you will observe **Security** mode is now set to **User's Identity.**

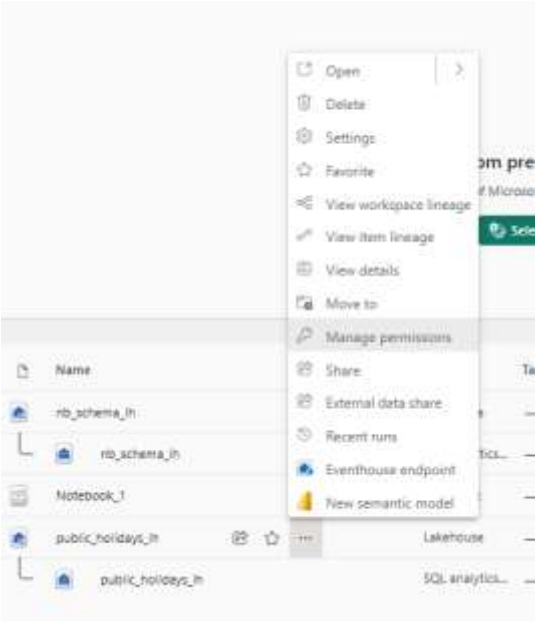


Checkpoint – at this stage, in your workspace ending in *_public*, you should have a lakehouse public_holidays_lh where you have opted-in for OneLake security.

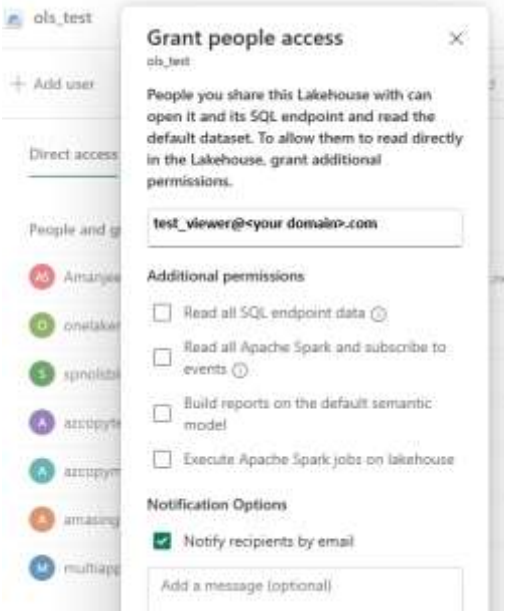
Step 3 – Grant permission to connect to lakehouse item

In Fabric, item **Read** permission is scoped to a single item and allows users to connect to the item and read its metadata.

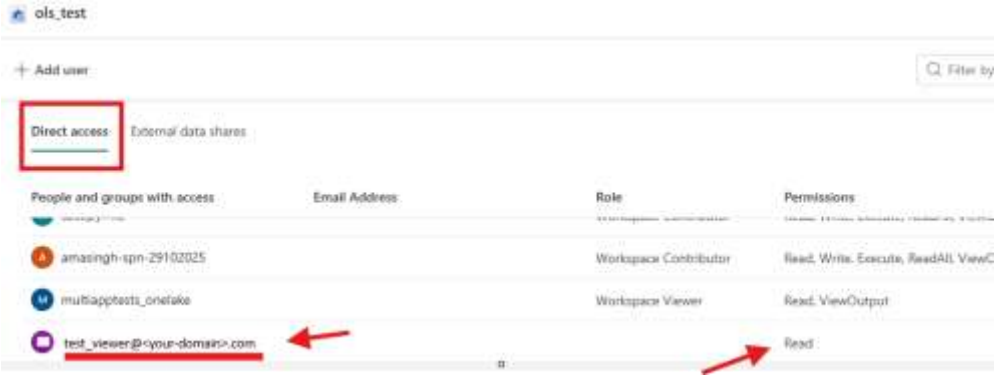
12. Switch to workspace view and then expand options against **public_holidays_lh** lakehouse by clicking ellipses **(...)** > **Manage Permissions**.



13. Select **+ Add User** and then enter **test_viewer**, and then select **Grant.** Ensure that you all **additional permissions** unchecked. By not selecting any permissions you are granting Read permission which allows a user to connect to an item without accessing any data.



14. You should now see **test_viewer** listed under **Direct Access** grants with **Read** permission.

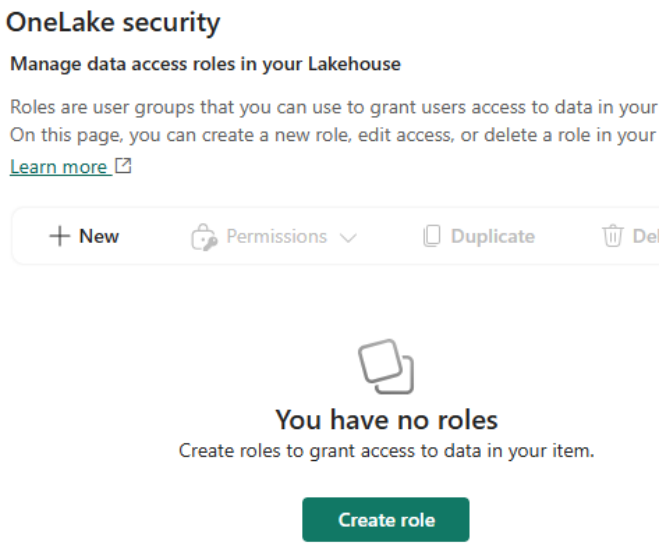


Step 4 – Create and configure a OneLake security role

In this exercise, you will configure a column-level and row-level security policies for **test_viewer** and verify that only allowed columns are visible when signed in as that user.

15. Select **Manage OneLake security** from lakehouse experience.

16. At this stage, there shouldn't be any roles listed. Select **Create Role.**

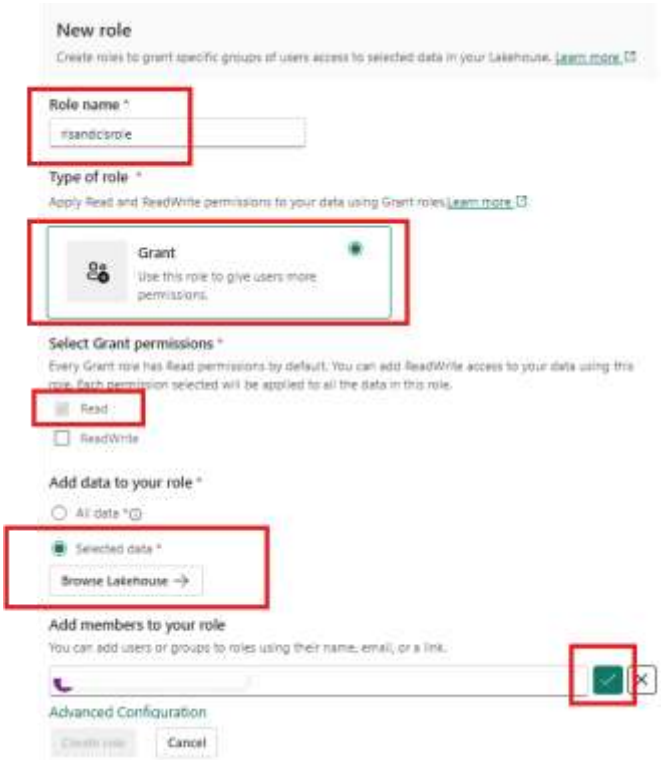


17. Select **Create Role** and fill out the details as shown here in the screenshot below.
    - **Role Name:** rlsandclsrole
      Note: You can use any name for your environment.

    - **Grant:** Select the radio button next to **Grant.** This will reveal Read and ReadWrite permissions. In this case, leave this with default selection.

    - **Add Data to your Role:** Choose the option **Selected Data** and then select **Browse Lakehouse.**

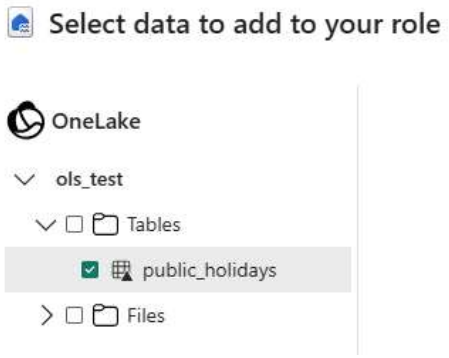
Note: You must first create a role before you can assign RLS or CLS policies.



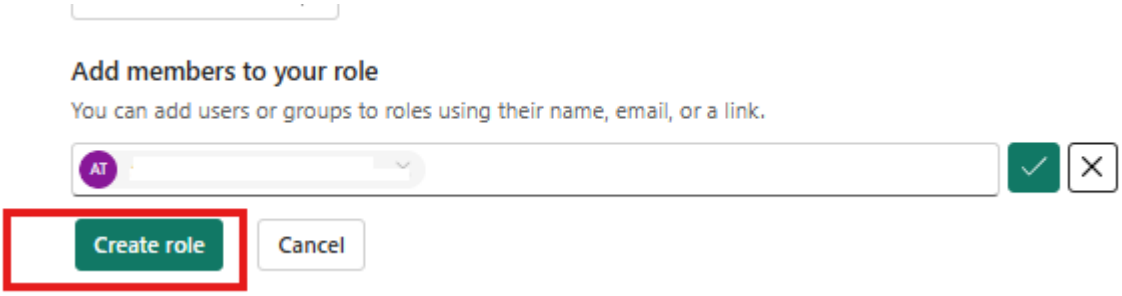
Note: Ensure that you click on green tick next to **Add members to your role** textbox.

18. Select **Brows Lakehouse** to access table listing. In the experience, select **public_holidays** table.

19. Under **Browse Lakehouse** dialog box, select **public_holidays** table and then select **Add data.** This is the dataset which your role would apply to.



20. Once you return to **New Role** experience, click on **Create Role** at the bottom of the screen.

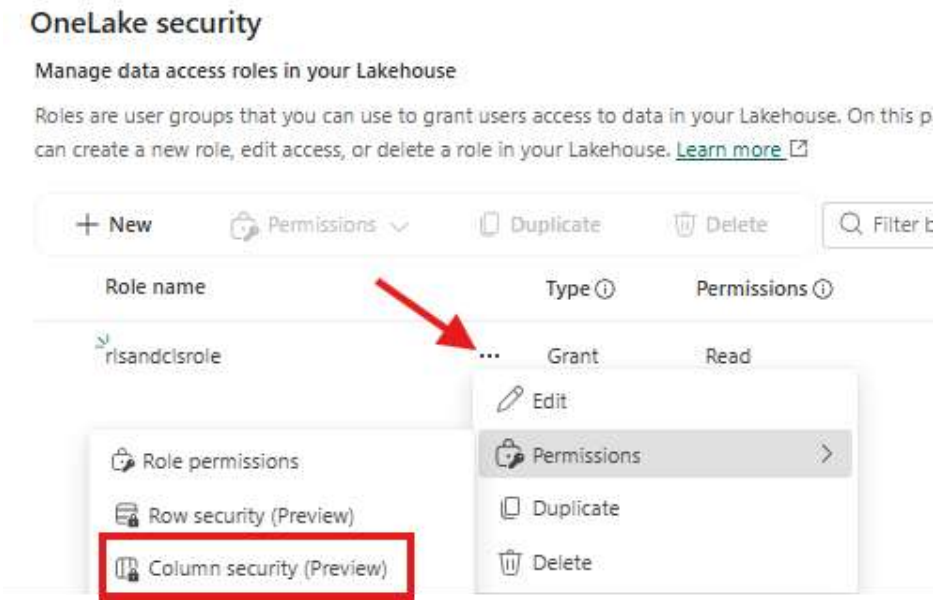


This completes role creation. The role defines access to the selected tables or folders and is assigned to the user **test_viewer**.

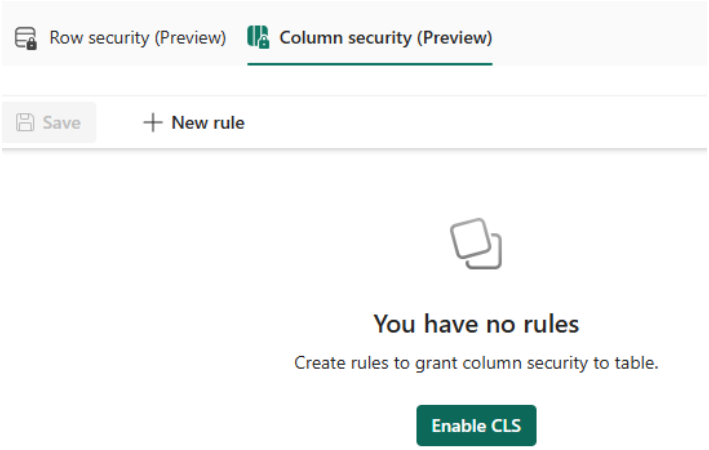
Step 5 – Configure Row-level (RLS) and Column-level (CLS) security policies and add them to a OneLake security role.

In this procedure, you will add RLS and CLS policies to the role created in **Step 4.**

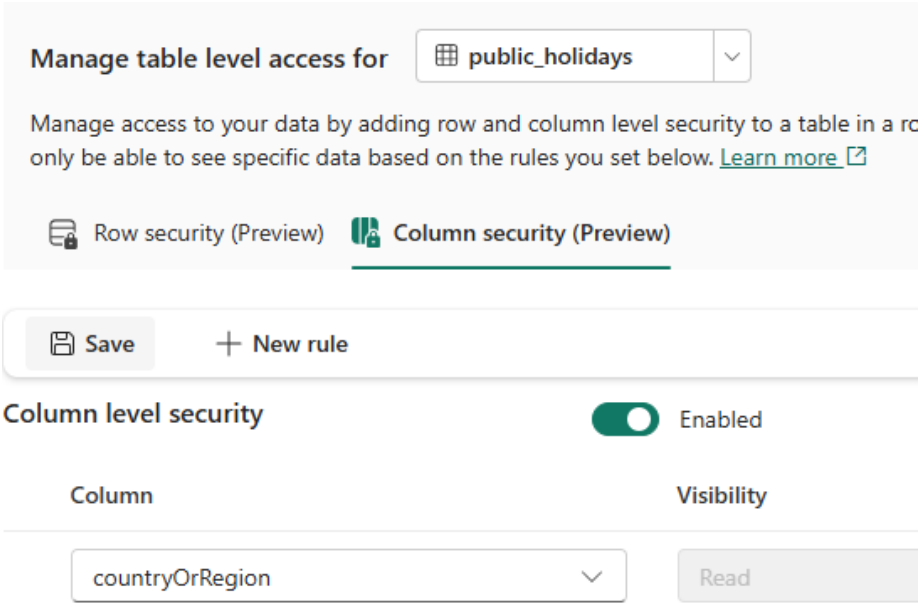
21. In the **OneLake security** experience, select the ellipsis (**...**) next to the **rlsandclsrole** role, then select **Column security**.



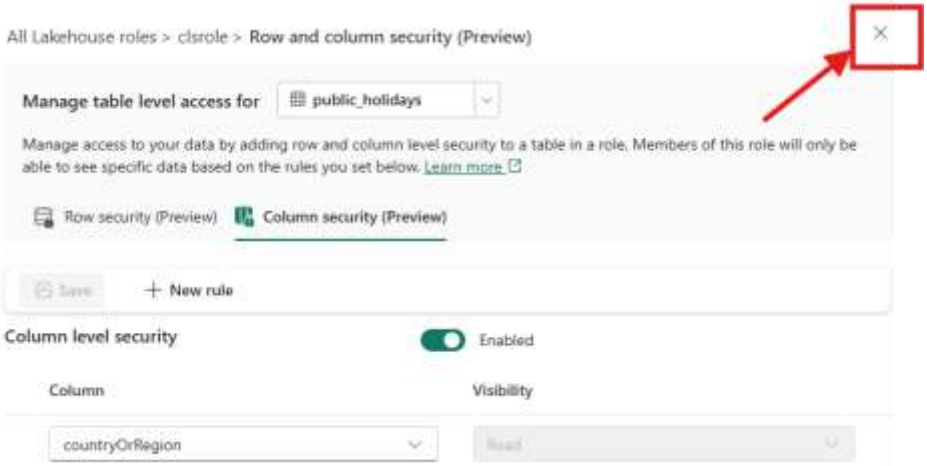
22. In the CLS experience, select **Enable CLS.**



23. Remove all columns and leave **countryOrRegion** and then select **Save.**

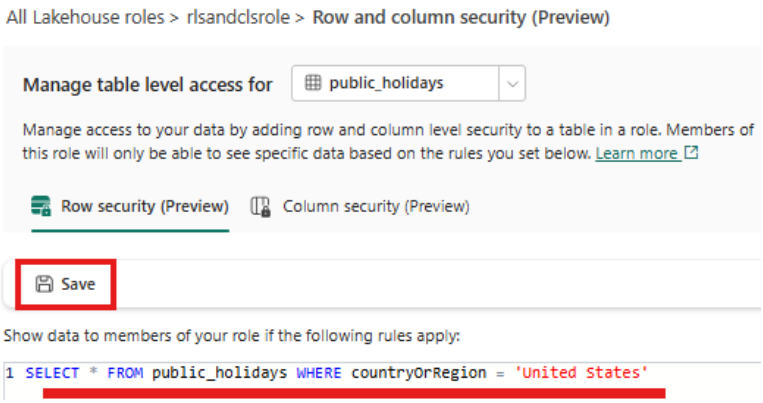


24. Once configured, OneLake security will return you to **Row and Column security** page. You can select **X** and close this window.

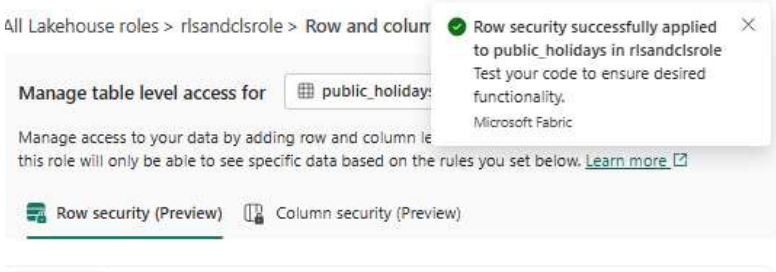


25. Select **Row security** tab – here, you will define row-level security policy which applies to this role.

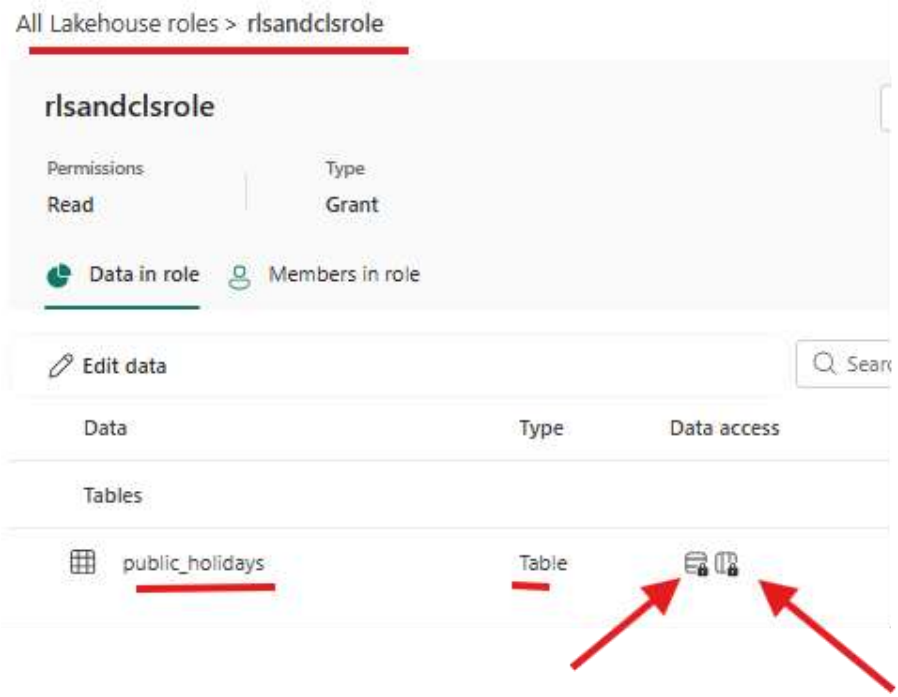
```
SELECT * FROM public_holidays WHERE countryOrRegion = 'United States'
```



26. Select **Save.** You should see a notification pop-up on the top right corner of the screen.



27. **Optional** – you can switch to OneLake security roles and see a summary of the policies which are part of this role.

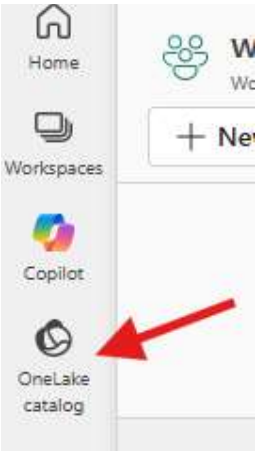


This completes configuration of a OneLake security role with a **row-level (RLS)** and a **column-level (CLS)** access policies.

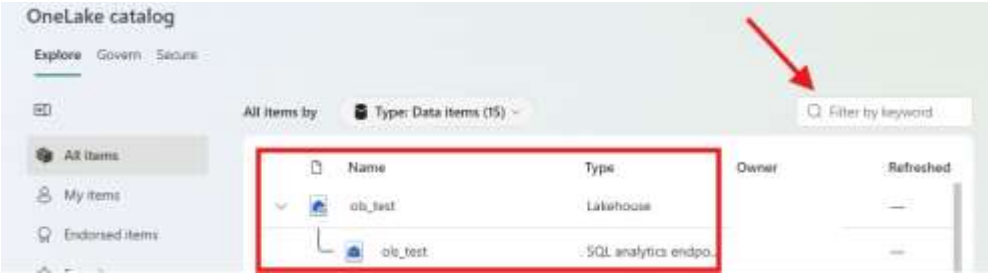
Step 6 – Test OneLake security role using SQL analytics endpoint

You will now test the OneLake security role assigned to test_viewer using SQL analytics endpoint and Data engineering (notebook) experiences.

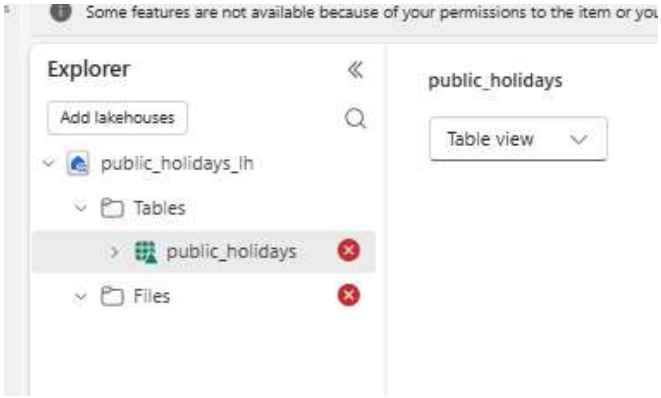
28. To test the permission, login as user **test_viewer**.
29. In the Fabric workspace – navigate to **OneLake catalog**.



30. Locate **ols_test** lakehouse. You can also use the search feature on the right side of the OneLake catalog experience.

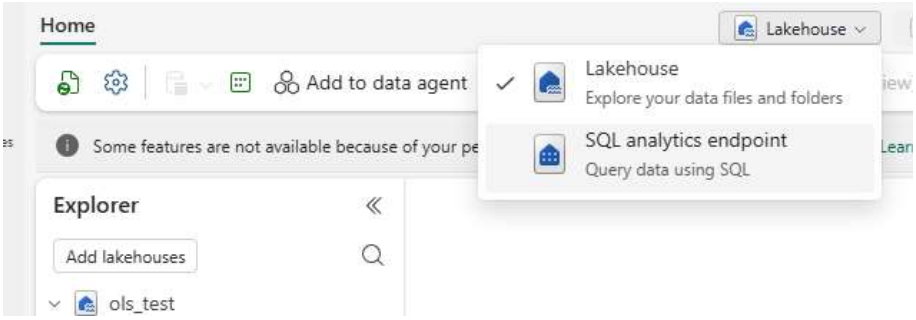


31. Open the lakehouse **public_holidays_lh**. In the explorer pane, you will observe that there's a **X** against lakehouse table **public_holidays** and **Files.**

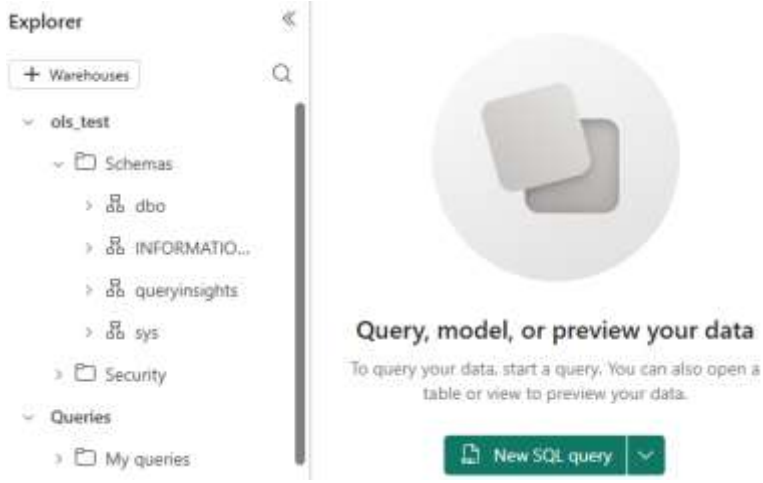


The preview window won't load and report an error as that uses SQL analytics endpoint and there's no mechanism to filter out rows (RLS) or columns (CLS) for the preview query. For now, you can ignore it.

32. Switch to **SQL analytics endpoint** experience.

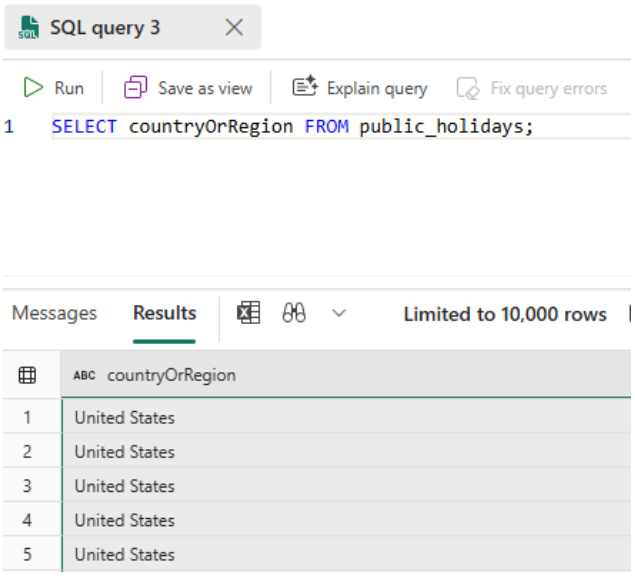


33. Open SQL editor by selecting **New SQL query.**

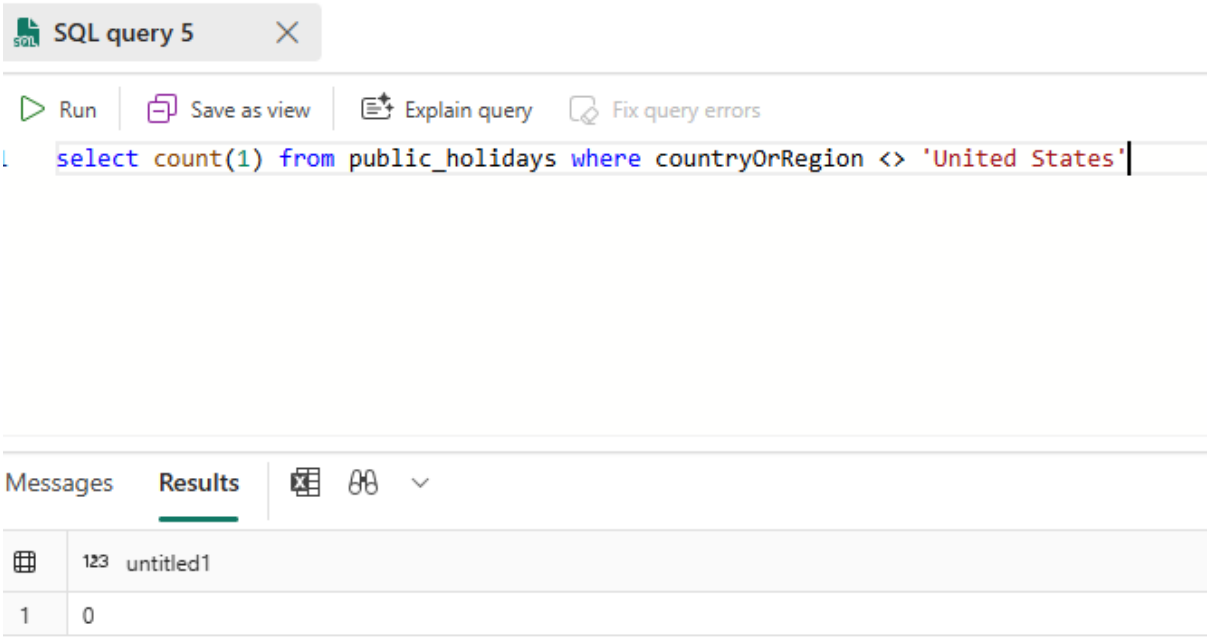


34. The CLS policy only allows one column **countryOrRegion**, so you can try the following query.

```
SELECT countryOrRegion FROM public_holidays;
```



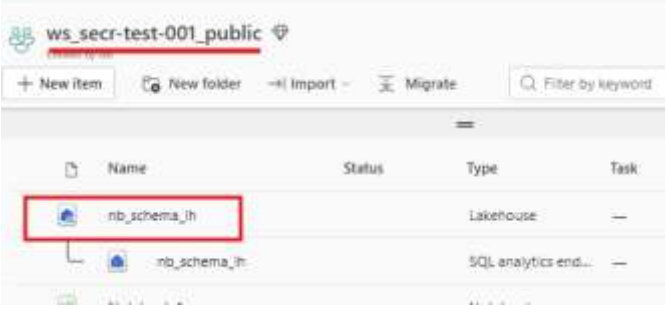
35. You can validate that RLS is enforced by running a query where you attempt to query records where predicate is **countryOrRegion <> 'United States'**.



Step 7 – Test OneLake security role using Fabric notebook

Additional steps are required for end users (viewers) to use notebooks successfully. Test_viewer here refers to the account you'll be using for testing the policies.

36. As the administrator (your username), navigate to workspace **ws_<*username*>_public** there will be a schema-enabled lakehouse **nb_schema_lh.**



This schema-enabled lakehouse is provisioned to demonstrate OneLake security using notebooks.

For Spark notebooks to successfully resolve table paths and honour OneLake security policies, notebook must be attached to a schema-enabled lakehouse, and workspace viewer access is required. These are temporary limitations and will be addressed soon.

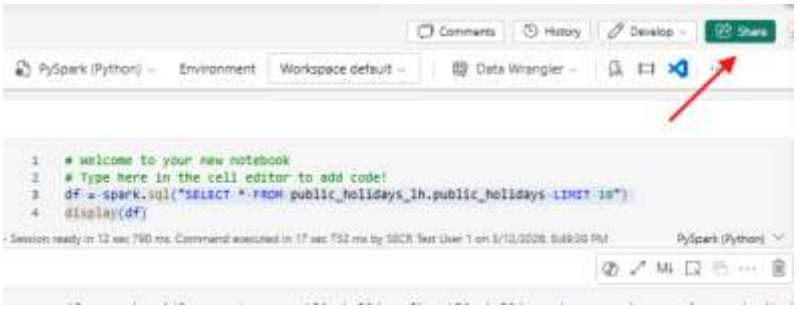
37. As a **Workspace Admin** in **ws_<username>_public**, create a notebook in the **nb_schema_lh** lakehouse. Use the code below to verify that **RLS** and **CLS** policies are enforced.

Ensure that you are using **Spark notebook** for this test.

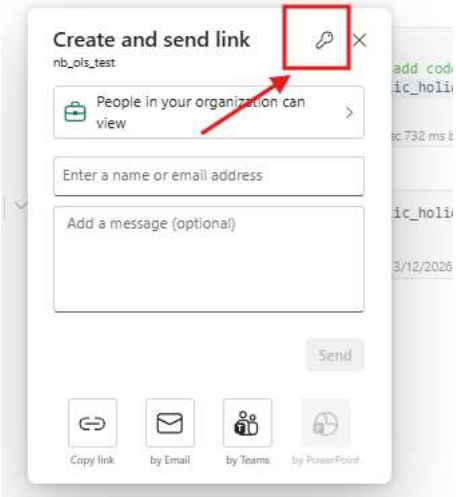
```
# Testing OneLake security CLS
df = spark.sql("SELECT * FROM public_holidays_lh.public_holidays LIMIT
10")
display(df)
```

```
# Testing OneLake security RLS
df = spark.sql("SELECT countryOrRegion FROM
public_holidays_lh.public_holidays where countryOrRegion <> 'United
States' LIMIT 10")
display(df)
```

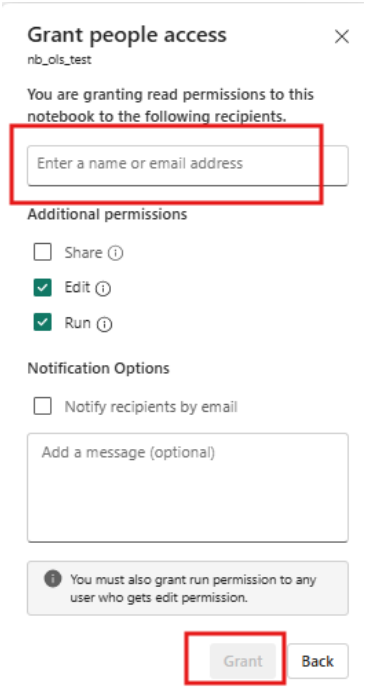
38. Once you have created the notebook, you must share it with the **test_viewer** with **Run** permissions. This will allow the viewer to run the code inside the notebook.



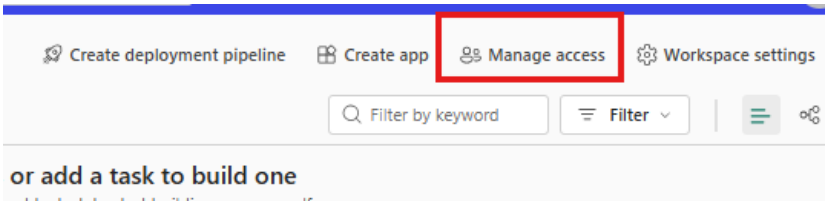
39. Click on the **key** symbol on top right corner.



40. In this case, we granted **Edit** and **Run** permissions to the viewer. In this case, we are using user account **secr-test-015** for testing view. You must enter username of the account you are using for testing and shared by the instructor.



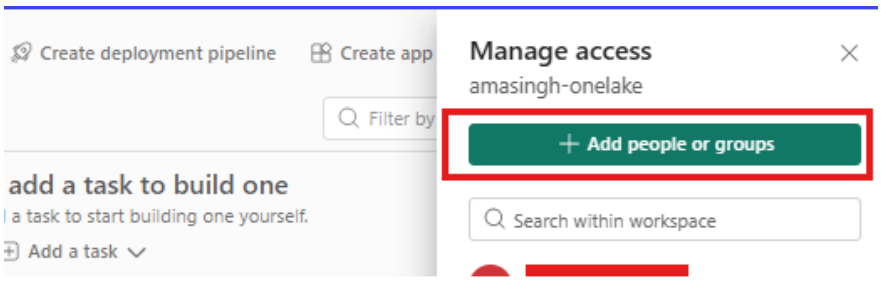
41. Navigate to **Manage Access** and assign workspace **Viewer** role to the test_viewer account you are using for validating OneLake security policies.



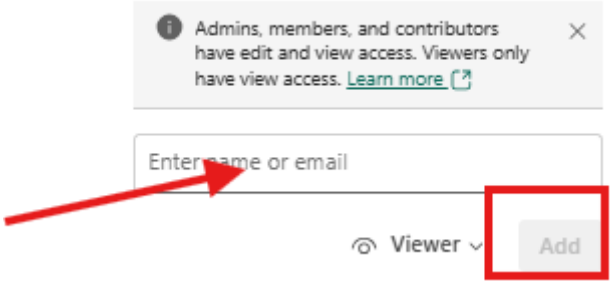
Both **workspace viewer** and **item permission** to run notebook are needed to run the notebook and test OneLake security policies.

Workspace roles and item permissions are control plane permissions, while OneLake security governs data plane access. Both are required for end users to run notebooks.

42. Select **Add people or groups.**

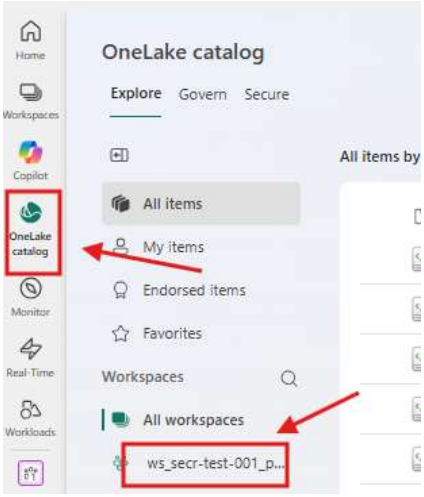


43. Add **test_viewer** in the text box and select **Add**. This will assign workspace viewer role to user test_viewer.

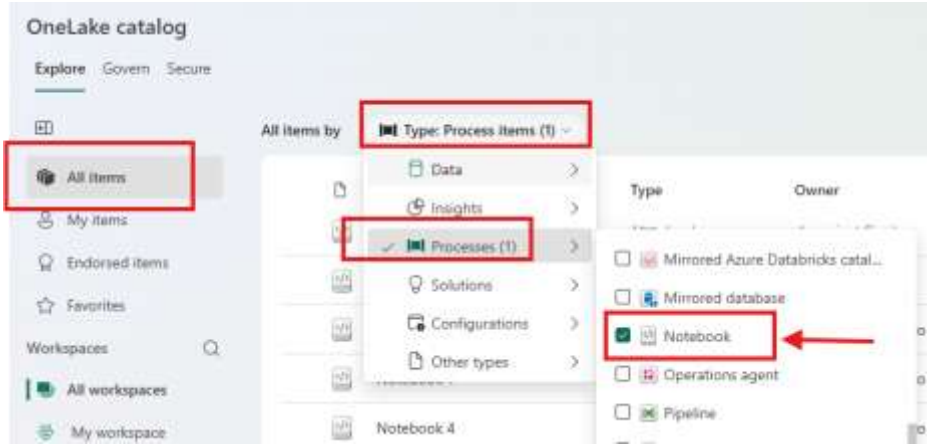


44. In a separate browser window, login as the **test_viewer** for testing OneLake security policies using notebook. If you are already logged in and have an existing session, please ensure that you refresh your session.

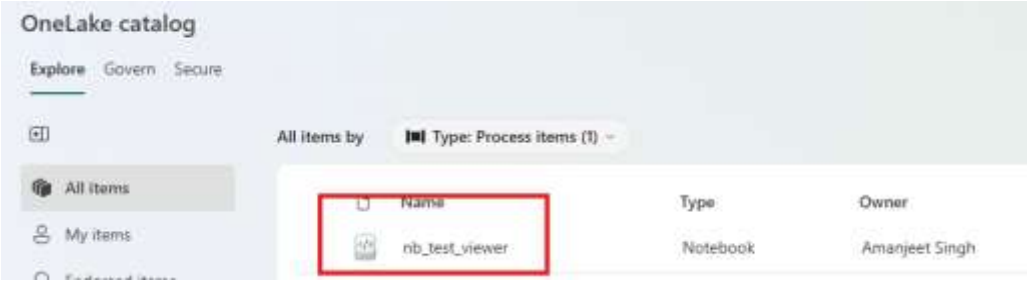
45. As **test_viewer**, navigate to **OneLake Catalog** and you should have your **ws_<*username*>_public** workspace visible. The workspace is visible/listed as you are logged in as **test_viewer** who has viewer access.



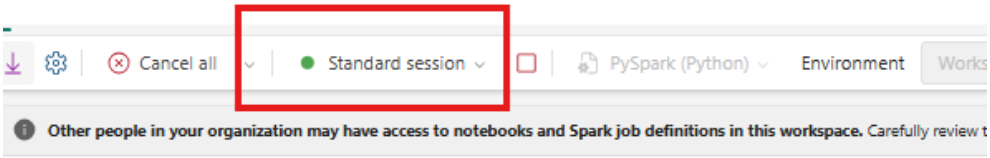
46. Set the filter to **Notebook** so that OneLake catalog displays all the notebooks. you can set the filter through these steps.



47. From the filtered results, select and select the notebook which has been provisioned for testing. In this case, name of the notebook is **nb_test_viewer**. You may have used a different notebook name.

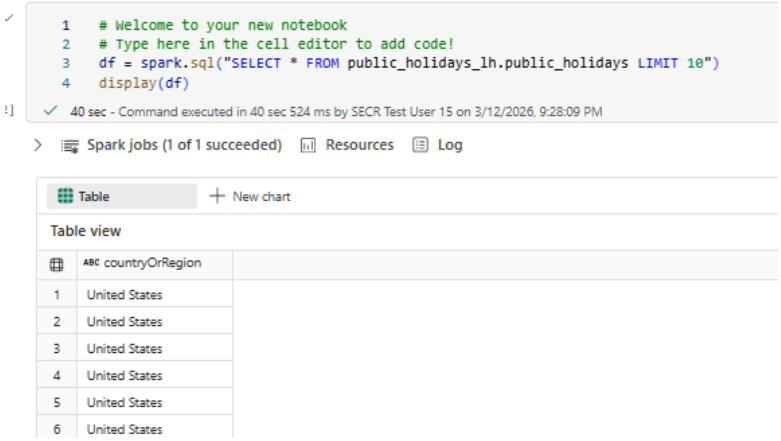


48. Connect the notebook to a **Standard session.**

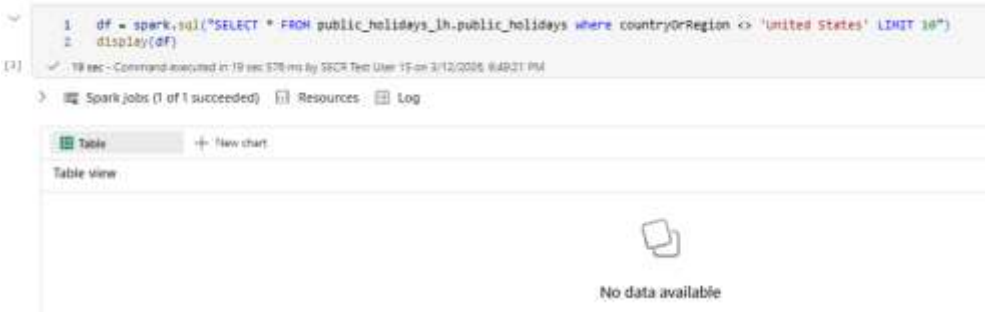


49. Since you are accessing the notebook as a viewer, you cannot edit it. You can run the cells which were populated earlier under **step 37**.

50. For **CLS** test – you will run the first cell which selects all columns. SparkSQL should only return values for **countryOrRegion** column as that's the only allowlisted column.



Run the next cell to validate **RLS enforcement** in the Spark notebook.
The query requests rows where **CountryOrRegion ≠ "United States"**. Since RLS restricts access to **CountryOrRegion = "United States"**, Spark returns **0 rows**, and the count is **0**.



Summary

In this module, you learnt how OneLake Security enforces RLS, and CLS to enforce least-privilege access across Fabric experiences (lakehouse and SQL analytics endpoint).

This concludes Module 4 – OneLake security.